02

User Workflow

The journey of a homeowner from one photo to a contractor-ready report: what they do at each step, what Miesign does in response, where the AI is involved, how long it takes, and what happens when something is not right.

1 Who uses Miesign and why
2 End-to-end flow with decisions
3 How steps unlock and resume
4 User and system, step by step
5 Home page
6 Step 01 Photo
7 Step 02 Parts
8 Step 03 Design (three ways)
9 Step 04 Preview
10 Steps 05 Estimate and 06 Report

PRODUCT
Miesign — AI exterior renovation planner and INR cost estimator

AUTHOR
Mihir Kapildev Tiwari

TRY IT
miesign.pages.dev — no sign-up

1 · USERS

Who uses Miesign and why

Primary: homeowners

Individual owners planning a repaint or facade renovation, and new owners before finishing. They cannot picture the result, are unsure which material suits which part, and get inconsistent quotations. Miesign lets them see their own house redesigned and walk into a contractor meeting with quantities and a costed breakdown.

Secondary: trade users

Contractors, architects, builders, real-estate consultants and material suppliers can use the same flow to show a client options quickly; the PDF report and the editable rates make it a discussion document rather than a final quote.

Design goals for the flow

- **One photo is enough** — no drawings, tape measures or accounts. An optional single measurement improves accuracy.
- **Never a dead end** — every rejection explains why and how to fix it; every AI result can be corrected by hand.
- **Plain language** — "Check what we found on your house", not "segmentation output". Numbers in ₹ with Indian grouping (₹1,99,538).
- **Work is never lost** — the project lives at its own link from the first upload; reopening it resumes where the user left off.

End-to-end flow with decisions

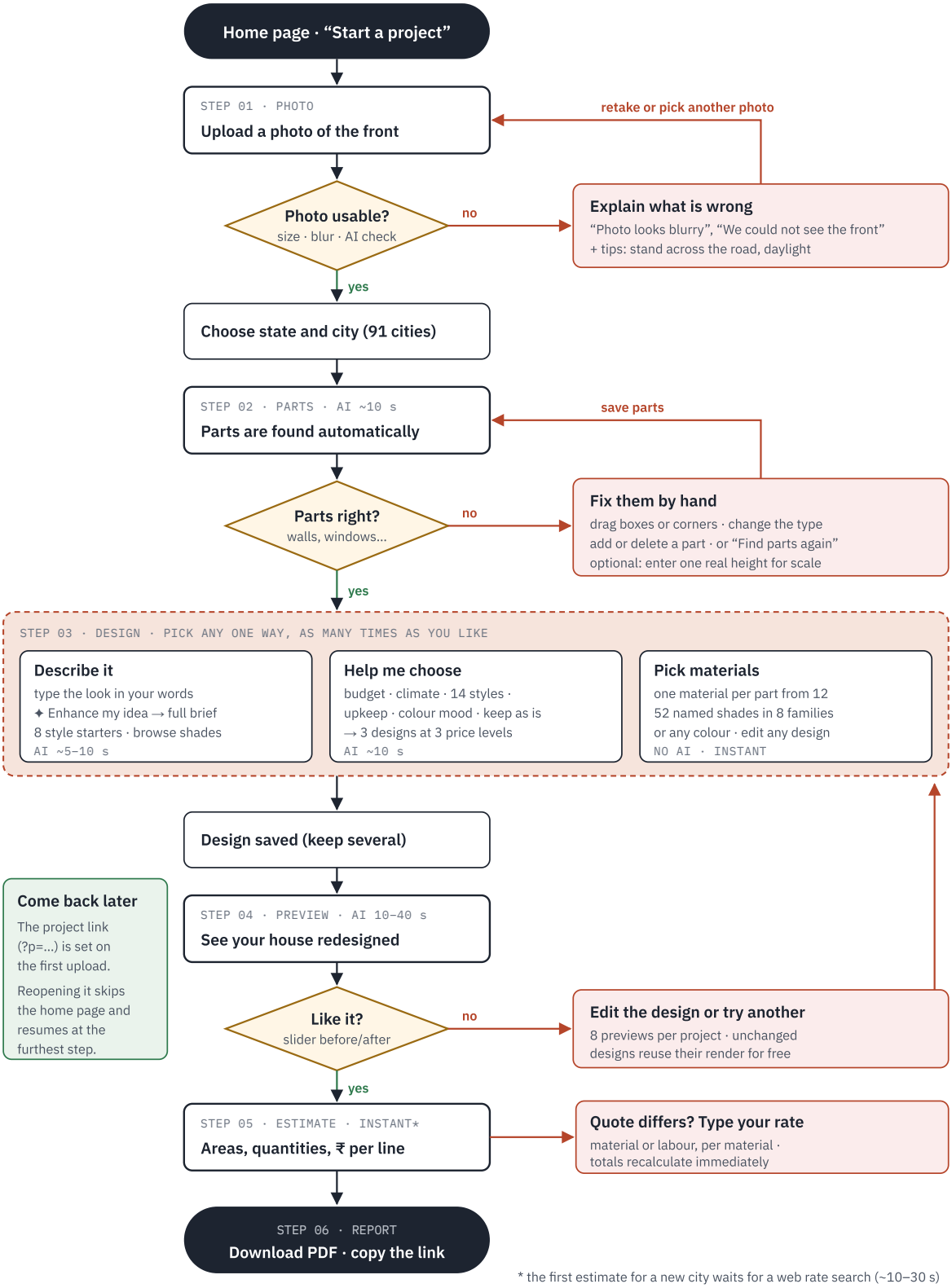


Fig 2.1 The full journey. Diamonds are decisions; red paths are the recovery loops that keep the user moving instead of failing.

3 · NAVIGATION

How steps unlock and resume

The six steps sit in a side rail on desktop and a compact bar with a step picker on phones. A step opens as soon as the data it needs exists, so users can jump back to any finished step without losing work.

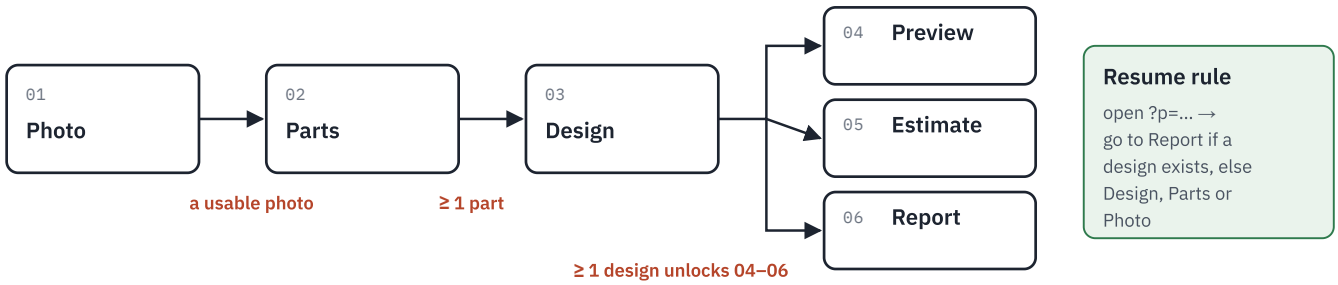


Fig 3.1 Step unlocking (`lib/steps.ts`). Labels on arrows are the conditions.

4 · RESPONSIBILITIES

User and system, step by step

Step	The user	Miesign	AI used	Typical time
Home	Reads what Miesign does, taps Start	Shows a real before/after, the six steps, what is included	—	—
01 Photo	Uploads 1–5 photos, picks state and city	Rotates, resizes, checks size and blur, then asks Gemini if a house front is visible; explains any rejection; first good photo becomes primary	Flash	4–6 s
02 Parts	Checks outlines, fixes any, optionally enters one real height	Detects 9 part types, adds a fallback wall, removes duplicate walls, sets a default scale	Flash	~10 s
03 Design	Describes, answers questions or picks materials and shades	Turns input into a validated design; the enhancer rewrites rough ideas; suggestions come at 3 price levels	Flash	5–10 s · manual instant
04 Preview	Renders, compares with the slider, iterates	Edits the photo with exact colours; 8 per project; refunds failures; caches unchanged designs	Flash Image	10–40 s
05 Estimate	Reads the breakdown, overrides rates	Measures parts, computes quantities with wastage and costs in ₹ with city rates	Flash + Search (rates, first time per city)	instant · 10–30 s new city
06 Report	Downloads the PDF, copies the link	Builds a PDF with both photos, materials, quantities, costs and assumptions	—	2–4 s

Home page

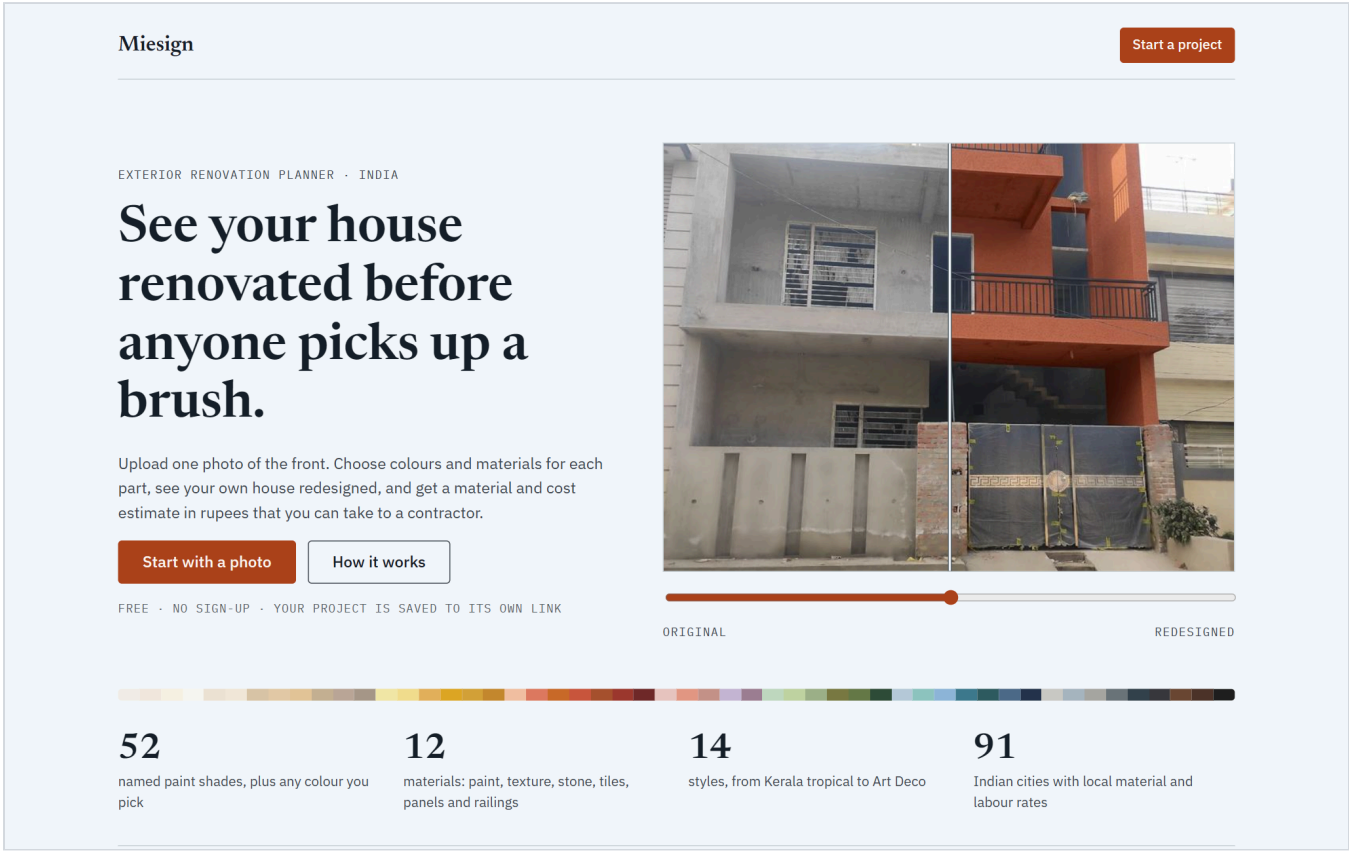


Fig 5.1 Home page on desktop: headline, a real before/after slider (Ludhiana), the 52-shade strip and key facts.

What the user sees

What Miesign does in one sentence, a working before/after slider on a real house, the numbers that matter (52 shades, 12 materials, 14 styles, 91 cities), the six steps, and a second real result. Two buttons lead in: “Start with a photo” and “Try it on your house”.

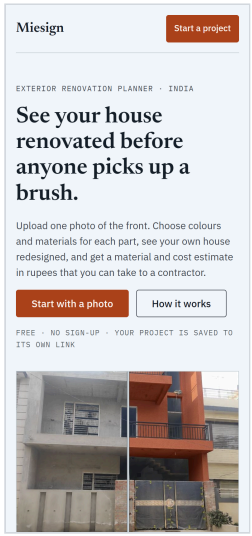


Fig 5.2 Same page at 390 px.

01 Photo

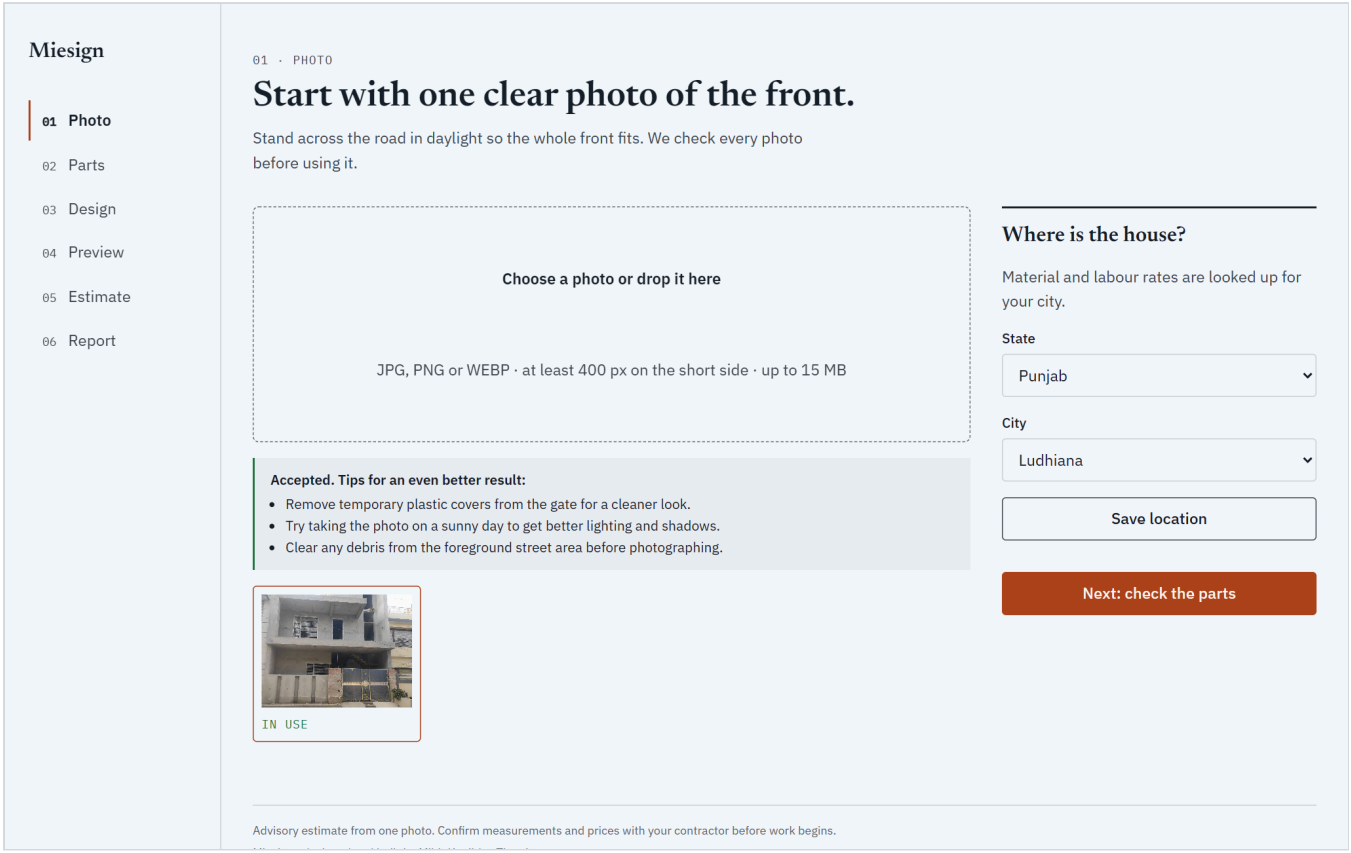


Fig 6.1 Photo step with an accepted photo, its tips and the city picker.

User does

- Taps or drops a JPG, PNG or WEBP (≥ 400 px short side, ≤ 15 MB).
- Picks state and city.

System does

- Applies EXIF rotation, resizes to ≤ 2048 px.
- Local size and blur check, then the Gemini house check.
- Creates the project link on the first upload.

When it is not right

- Rejections say why (“looks blurry”, “could not see the front”) with retake tips.
- Up to 5 photos; pick which one to use.

02 Parts

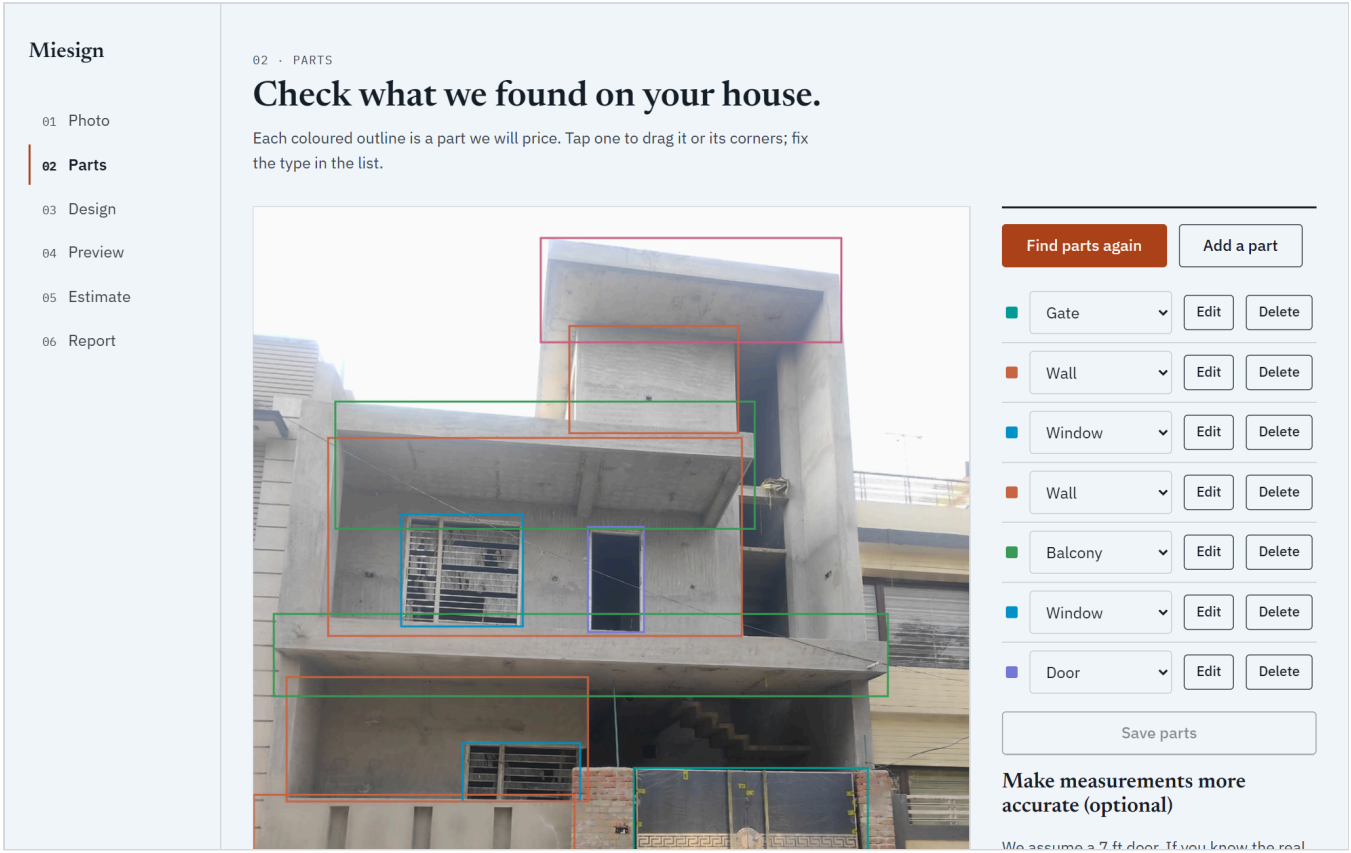


Fig 7.1 Detected parts drawn on the photo, one colour per type, with the editable list.

User does

- Checks each outline.
- Drags a box or its corners, changes a type, adds or deletes a part, saves.
- Optionally enters one real height (e.g. door = 7 ft).

System does

- Detection starts automatically with a fluid loader over the photo.
- Finds walls, windows, doors, balconies, pillars, parapets, gates, roof edges, railings.
- Sets a default scale from doors/windows.

When it is not right

- “Find parts again” replaces the set (asks first if edits are unsaved).
- A wall is always present; duplicate walls are removed.

03 Design — three ways in

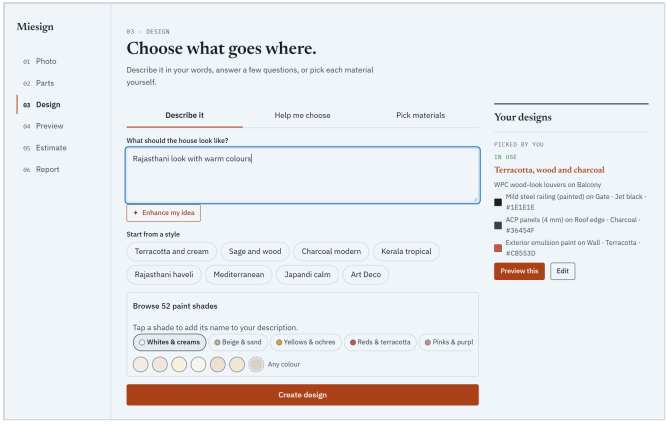


Fig 8.1 Describe it: free text, "Enhance my idea", style starters and the shade browser.

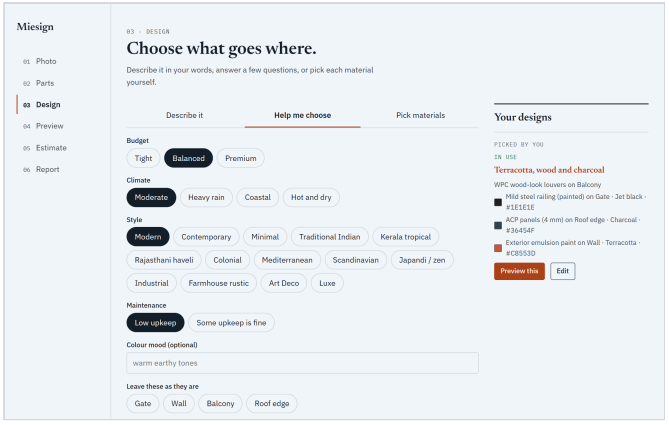


Fig 8.2 Help me choose: budget, climate, 14 styles, upkeep, colour mood.

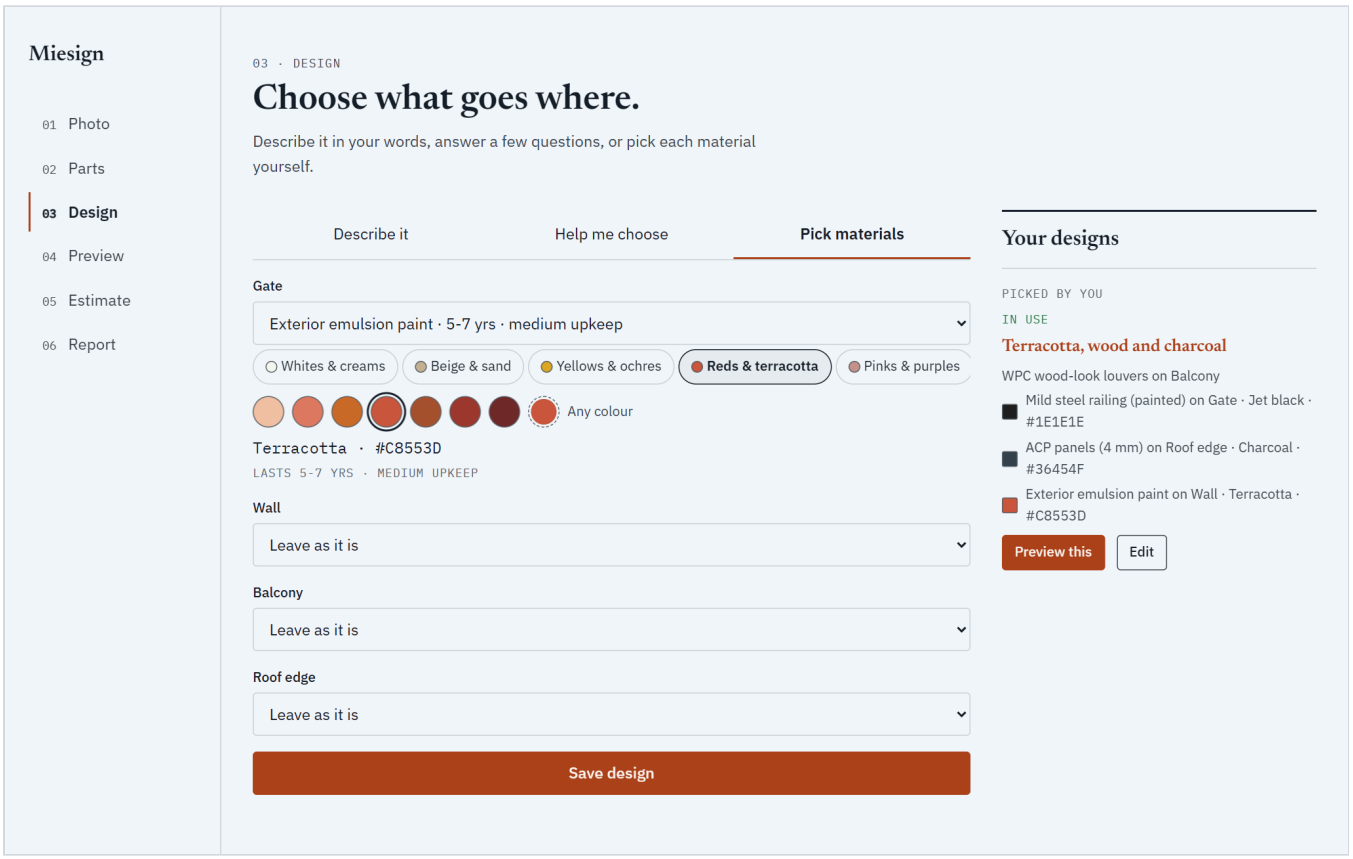


Fig 8.3 Pick materials: one material per part, then a shade from 8 families (here Reds & terracotta → Terracotta #C8553D) or any colour. The saved design appears on the right.

Describe it

Type the look in your own words. **Enhance my idea** rewrites a rough idea into a full brief using real material and shade names (with Undo). 8 style starters and a 52-shade browser help.

Help me choose

Five questions — budget, climate, style (14, from Kerala tropical to Art Deco), upkeep, colour mood — plus parts to keep. Returns three designs at three price levels with reasons.

Pick materials

A material per part from 12, filtered to what suits that part, showing durability and upkeep; then a named shade or any colour. Any saved design can be edited here.

04 Preview



Fig 9.1 While rendering: the fluid loader in the design's colours.



Fig 9.2 Result: drag the slider between original and redesigned.

User does

- Picks a design, taps Render.
- Drags the slider; reads “What changes”.
- Goes back to edit, or on to the cost.

System does

- Sends the photo, a JSON prompt and a swatch card of exact colours to the image model.
- Keeps structure, windows and doors as they are.

Limits

- 8 previews per project, shown as “2 of 8 used”.
- Failed renders are refunded; unchanged designs reuse their render.

05 Estimate and 06 Report

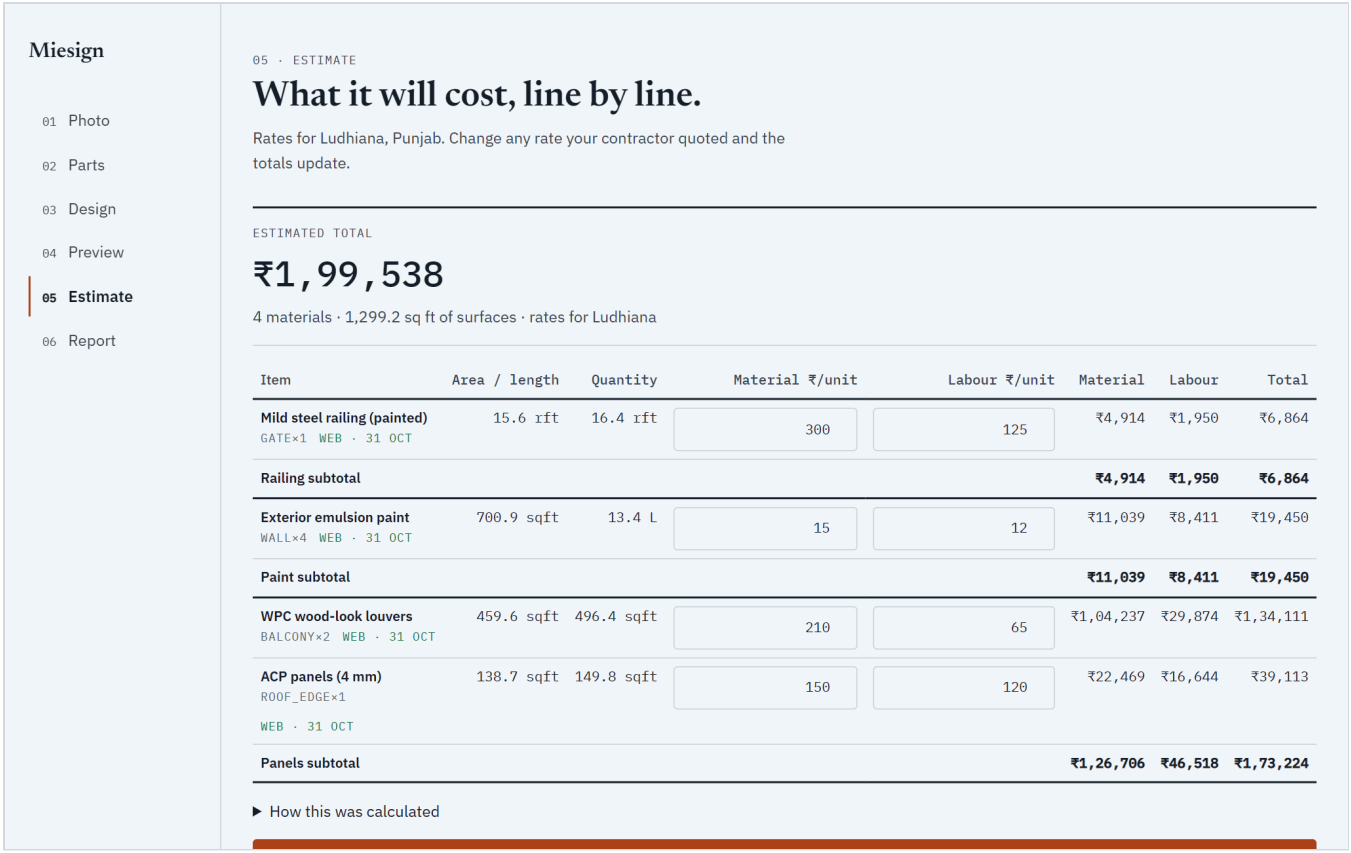


Fig 10.1 Estimate: total first, then each material with area, quantity with wastage, editable material and labour rates, and category subtotals.

Estimate

Every line shows which parts it covers (e.g. WALL×4), where its rate came from (WEB with date, ESTIMATE, or YOURS), and can be changed. “How this was calculated” lists the assumptions and the web sources for the rates.

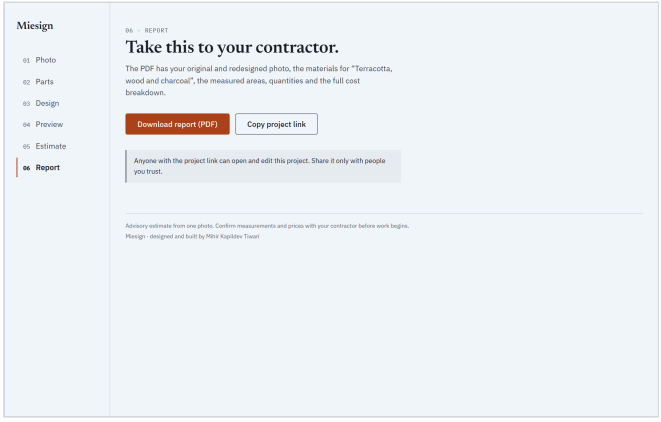


Fig 10.2 Report: download the PDF or copy the project link.